

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech. **Branch :** Mechanical Engineering/Mechanical Engineering(Sandwich)
Semester : IV Subject Code and Name: BTMC401/BTMEC401_Y19 Manufacturing Processes-I

Max Marks: 60

Date:19/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Question : Expendable moulds are typically made of----- a. sand b. plaster c. ceramics d. all of the above	Level 1	1
2	Question : True centrifugal casting can be used for manufacturing of -- a. gun barrels b. connecting rods c. crank shafts d. none of the above	Level 1	1
3	Question :The hot-chamber die-casting can be used for casting of the following material(s)----- a. zinc b. magnesium c. tin d. all of the above	Level 1	1
4	Question : Generally, the thickness of the sheets is in the order of ----- a. less than 6 mm b. greater than 6 mm c. equal to 10 mm d. none of the above	Level 1	1
5	Question :The rolled strip will be thicker at its centre than at its edges, known as----- a. crown b. camber c. chamber d. all of the above	Level 1	1
6	Question :In oxy-acetylene gas welding, if the oxygen is insufficient for full combustion, the flame is known as	Level 1	1

	a. redusing or carburizing	b. oxydizing	c. neutral flame	d. none of the above		
7	Question : In Gas Tungsten-arc welding (TIG), the type of electrode used is-----				Level 1	1
	a. consumable-electrode	b.nonconsumable electrode	c. consumable and and nonconsumable	d. none of the above		
8	Question :Porosity in welds may be caused by				Level 1	1
	a. gases released during melting of the weld area	b. chemical reactions during welding	c. contaminants	d. all of the above		
9	Question : Which of the following material(s) can be easily soldered				Level 1	1
	a. copper	b. silver	c. gold	d. all of the above		
10	Question : Which of the following part(s) that can be manufactured using turning process				Level 1	1
	a. grooved parts	b. shafts	c. spindles	d. all of the above		
11	Question : The self-centering can be achieved using the following----				Level 1	1
	a. three-jaw chuck	b. four-jaw chuck	c. five-jaw chuck	d. none of the above		
12	Question : Broaching process can be used to produce the following-----				Level 1	1
	a. circular holes	b. square holes	c. irregular sections	d. all of the above		
Q. 2	Solve the following.					12
A)	What are the different types of patterns are used in sand casting? Explain any three.				CO1	6
B)	Explain the investment casting process with a neat sketch. How the individual wax patterns are attached on a ‘tree’ in investment casting?				CO1	6

Q.3	Solve the following.		12
A)	Explain any three rolling mill stand arrangements.	CO2	6
B)	Compare and differentiate extrusion with rolling and forging.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What is the principle of welding? How the filler metal and parent metal get diffused, explain?	CO3	6
B)	Explain how cutting takes place when an oxyfuel-gas torch is used?	CO3	6
C)	How the welding is done in Metal Inert-Gas (MIG) welding process?	CO3	6
Q.5	Solve Any Two of the following.		12
A)	What is turning? What kind of chips are produced by turning? Which type of chips are desirable?	CO4	6
B)	Explain how external threads are cut on a lathe?	CO4	6
C)	What is the difference between turret lathe and capstan lathe? Explain it with a neat sketch.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Describe the different milling operations with neat sketches (any three)?	CO5	6
B)	Define broaching. Explain the geometric features of broach with their functions.	CO5	6
C)	What are the different processes used for finishing of gears? Explain any three processes.	CO6	6
	*** End ***		